Unit 3

Presented by : prof.Kundan Makwana

Data Link Layer

- The Data Link Layer is the second layer of the OSI (Open Systems Interconnection) model. It acts as a bridge between the Physical Layer (which transmits raw bits) and the Network Layer (which handles logical addressing and routing). The primary function of the Data Link Layer is to ensure reliable data transfer between two directly connected nodes.
- The Data Link Layer is further divided into two sublayers:
- LLC (Logical Link Control) : Error checking and frame synchronization
- MAC (Media Access Control) : Physical addressing and medium access control

- Functions of the Data Link Layer
- Before going into applications, it's important to understand what are the functions of Data Link Layer :
- Framing: Divides data into manageable units called frames.
- Error Detection and Correction: Detects and sometimes corrects errors using techniques like CRC (Cyclic Redundancy Check).
- Flow Control: Ensures that the sender does not overwhelm the receiver.
- MAC (Media Access Control): Determines how devices on a network share the communication medium.
- Addressing: Uses MAC addresses to identify devices on the same network.

- Logical Link Control Sub-layer (LLC) -
- Provides the logic for the data link, Thus it controls the synchronization, flow control, and error checking functions of the data link layer. Functions are -
- (i) Error Recovery.
- (ii) It performs the flow control operations.
- (iii) User addressing.

- Media Access Control Sub-layer (MAC) -
  - It is the second sub-layer of data-link layer. It controls the flow and multiplexing for transmission medium. Transmission of data packets is controlled by this layer. This layer is responsible for sending the data over the network interface card.
- Functions are -
- (i) To perform the control of access to media.
- (ii) It performs the unique addressing to stations directly connected to LAN.
- (iii) Detection of errors.

- Functions of the Data Link Layer
- Before going into applications, it's important to understand what are the functions of Data Link Layer :
  - Framing: Divides data into manageable units called frames.
  - Error Detection and Correction: Detects and sometimes corrects errors using techniques like CRC (Cyclic Redundancy Check).
  - Flow Control: Ensures that the sender does not overwhelm the receiver.
  - MAC (Media Access Control): Determines how devices on a network share the communication medium.
  - Addressing: Uses MAC addresses to identify devices on the same network.

Design Issues in Data Link Layer

- 1. Services provided to the network layer -
- The data link layer act as a service interface to the network layer. The principle service is transferring data from network layer on sending machine to the network layer on destination machine. This transfer also takes place via DLL (Data link-layer).
- It provides three types of services:
- Unacknowlwdged and connectionless services.
- Acknowledged and connectionless services.
- Acknowledged and connection-oriented services

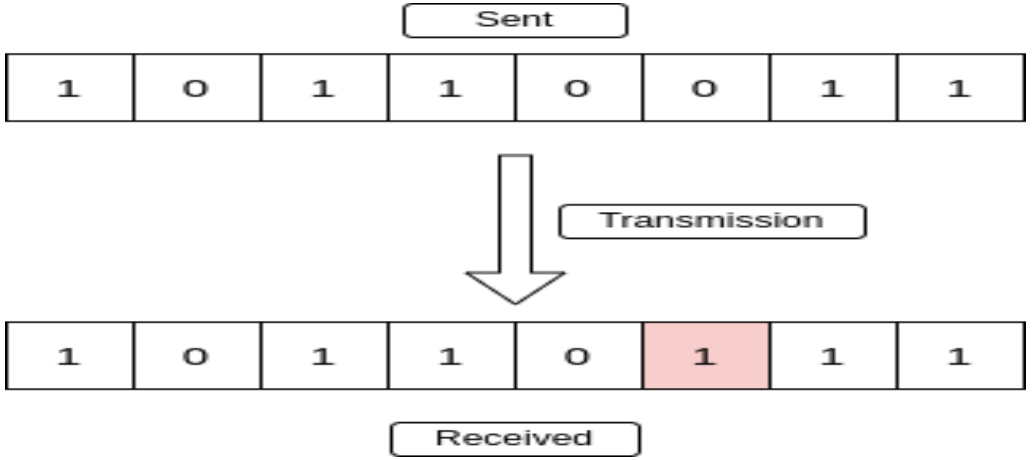
- Unacknowledged and connectionless services:
  - Here the sender machine sends the independent frames without any acknowledgement from the sender.
  - There is no logical connection established.
- Acknowledged and connectionless services:
  - There is no logical connection between sender and receiver established.
  - Each frame is acknowledged by the receiver.
  - If the frame didn’t reach the receiver in a specific time interval it has to be sent again.
  - It is very useful in wireless systems.

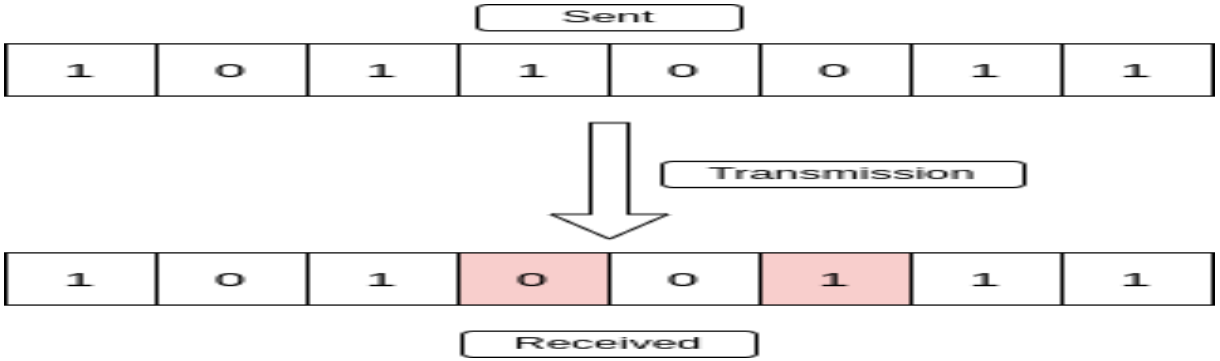
- 2. Frame synchronization -
  - The source machine sends data in the form of blocks called frames to the destination machine. The starting and ending of each frame should be identified so that the frame can be recognized by the destination machine.
- 3. Flow control -
  - Flow control is done to prevent the flow of data frame at the receiver end. The source machine must not send data frames at a rate faster than the capacity of destination machine to accept them.
- 4. Error control -
  - Error control is done to prevent duplication of frames. The errors introduced during transmission from source to destination machines must be detected and corrected at the destination machine.

Error Detection

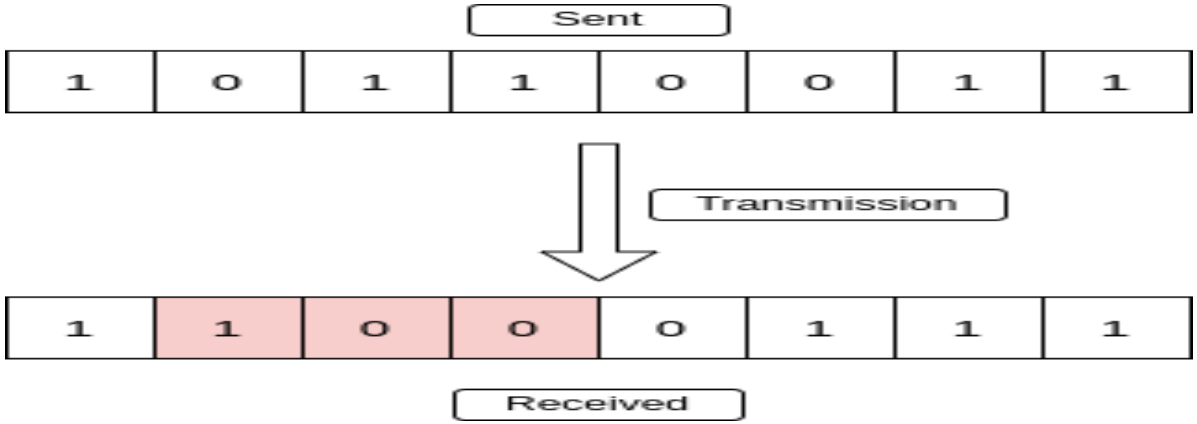
- Error detection in computer networks ensures that data is transmitted accurately without corruption. It works by adding extra (redundant) bits to data so the receiver can verify its correctness. If errors are found, the receiver discards the frame and requests retransmission.
- Detects errors caused by noise or interference.
- Uses redundant bits for consistency checking.
- Common methods: Parity Check, Checksum, CRC.
- Ensures data integrity and reliable communication.

Types of Errors

- 1. Single-Bit Error
- A single-bit error refers to a type of data transmission error that occurs when one bit (i.e., a single binary digit) of a transmitted data unit is altered during transmission, resulting in an incorrect or corrupted data unit.
- 

- 2. Multiple-Bit Error
- A multiple-bit error is an error type that arises when more than one bit in a data transmission is affected. Although multiple-bit errors are relatively rare when compared to single-bit errors, they can still occur, particularly in high-noise or high-interference digital environments.
- 

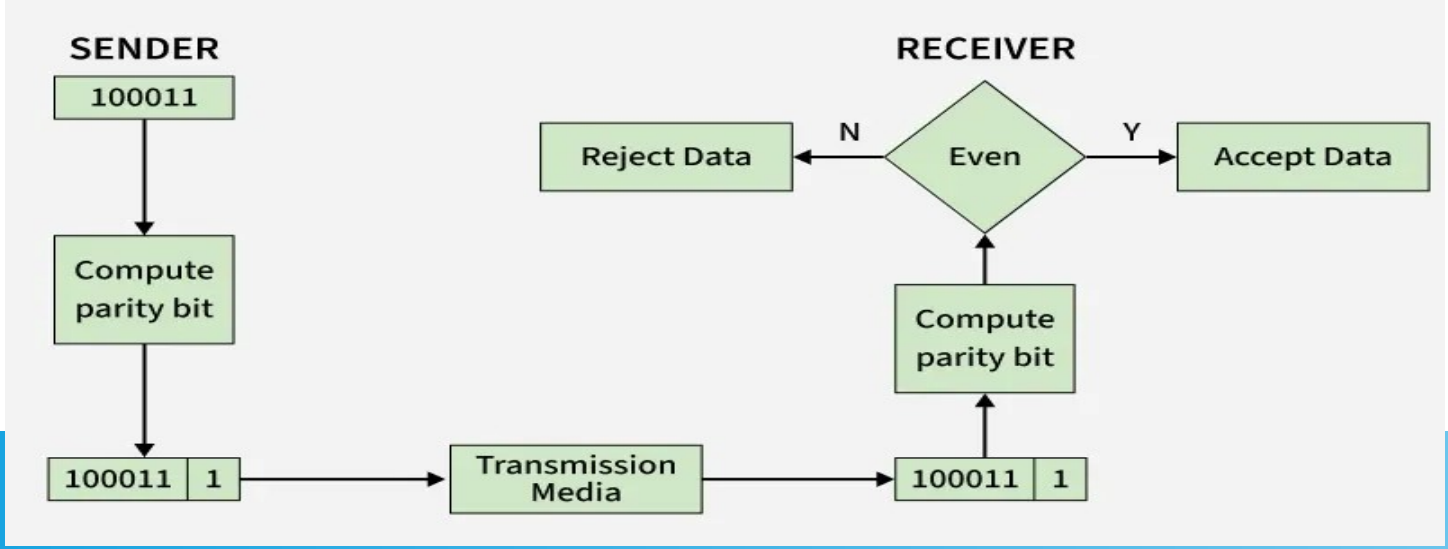
- 3. Burst Error
- When several consecutive bits are flipped mistakenly in digital transmission, it creates a burst error. This error causes a sequence of consecutive incorrect values.
-



Error Detection Methods

- To detect errors, a common technique is to introduce redundancy bits that provide additional information. Various techniques for error detection include:
- Simple Parity Check
- Two-Dimensional Parity Check
- Checksum
- Cyclic Redundancy Check (CRC)

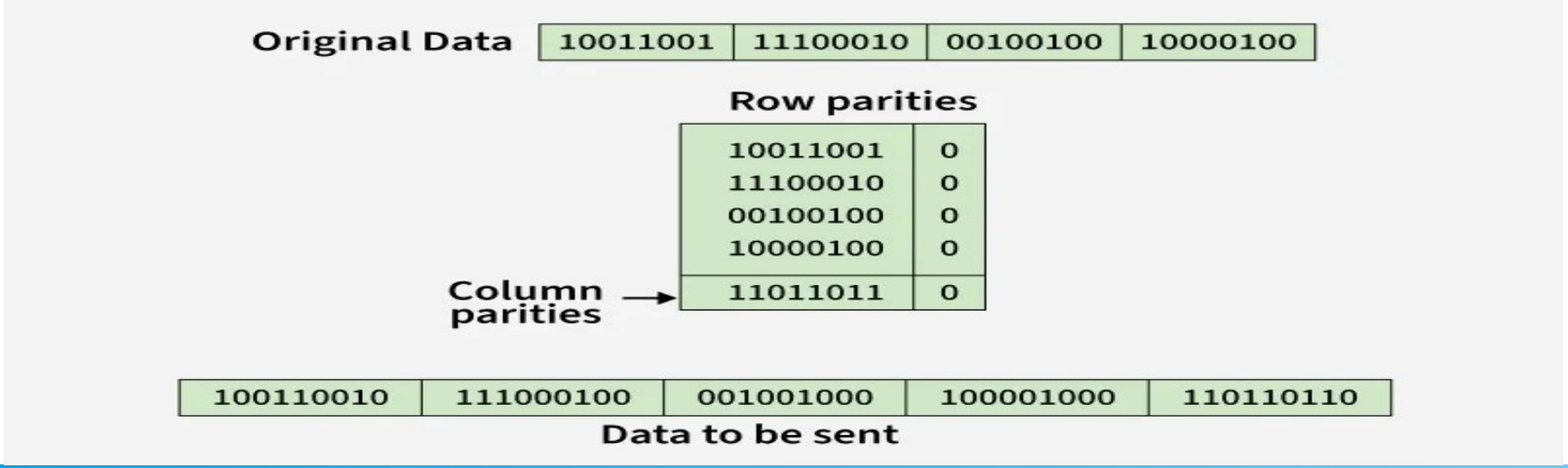
- 1. Simple Parity Check
- Simple bit parity is a simple error detection method that involves adding an extra bit to a data transmission. It works as:
  - 1 is added to the block if it contains an odd number of 1's, and
  - 0 is added if it contains an even number of 1's



- Advantages of Simple Parity Check
  - Simple parity check can detect all single bit error.
  - Simple parity check can detect an odd number of errors.
  - Implementation: Simple Parity Check is easy to implement in both hardware and software.
  - Minimal Extra Data: Only one additional bit (the parity bit) is added per data unit (e.g., per byte).
  - Fast Error Detection: The process of calculating and checking the parity bit is quick, which allows for rapid error detection without significant delay in data processing or communication.
  - Single-Bit Error Detection: It can effectively detect single-bit errors within a data unit, providing a basic level of error detection for relatively low-error environments.

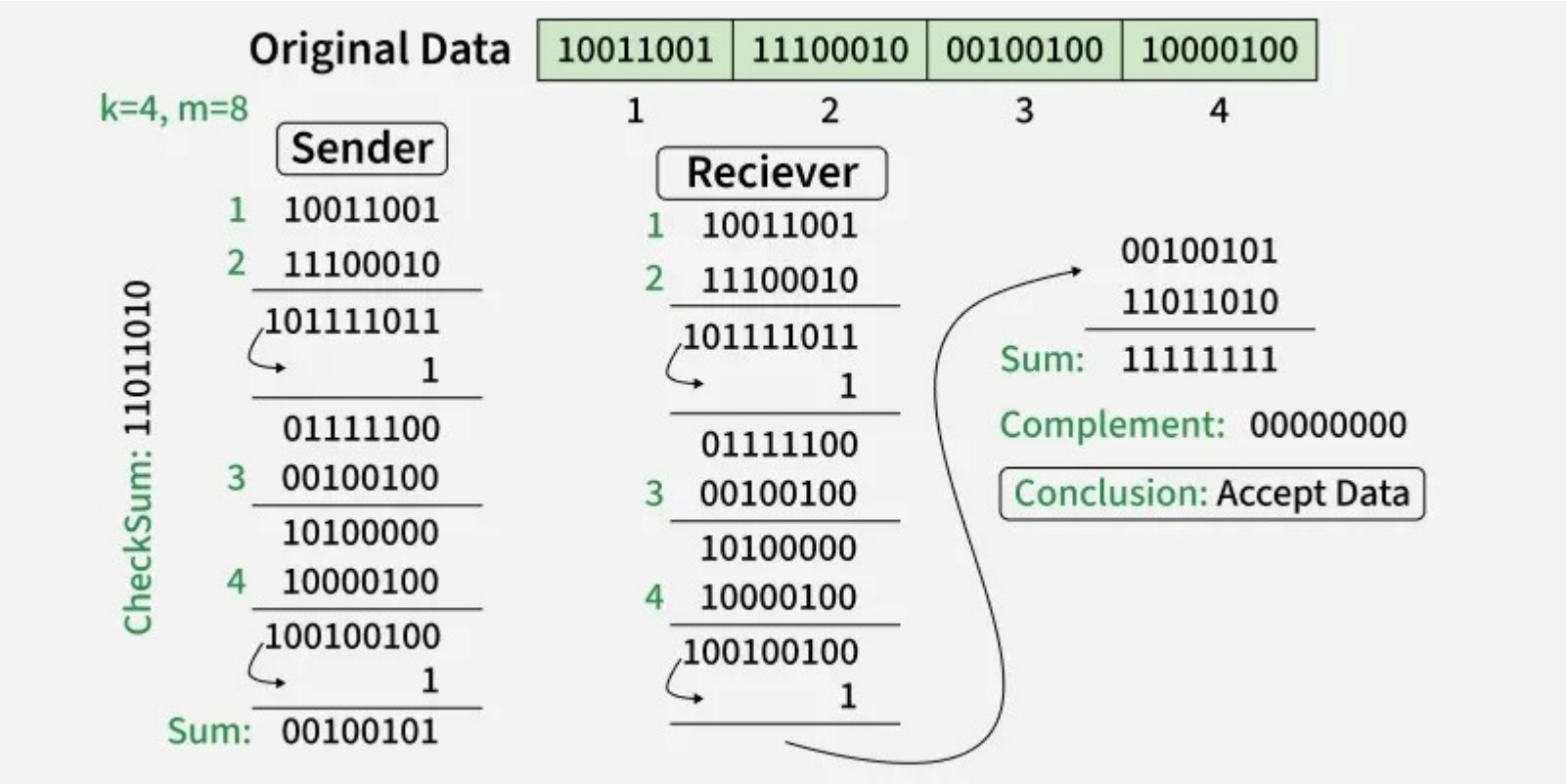
- Disadvantages of Simple Parity Check
  - Single Parity check is not able to detect even no. of bit error.
  - For example: the Data to be transmitted is 101010. Codeword transmitted to the receiver is 1010101 (we have used even parity).
  - Let's assume that during transmission, two of the bits of code word flipped to 1111101.
  - On receiving the code word, the receiver finds the no. of ones to be even and hence no error, which is a wrong assumption.

- 2. Two-Dimensional Parity Check
  - Two-dimensional Parity check bits are calculated for each row, which is equivalent to a simple parity check bit. Parity check bits are also calculated for all columns, then both are sent along with the data. At the receiving end, these are compared with the parity bits calculated on the received data.



- Advantages of Two-Dimensional Parity Check
  - Two-Dimensional Parity Check can detect and correct all single bit error.
  - Two-Dimensional Parity Check can detect two or three bit error that occur any where in the matrix.
- Disadvantages of Two-Dimensional Parity Check
  - Two-Dimensional Parity Check can not correct two or three bit error. It can only detect two or three bit error.
  - If we have a error in the parity bit then this scheme will not work.

- 3. Checksum
  - Checksum error detection is a method used to identify errors in transmitted data. The process involves dividing the data into equally sized segments and using a 1's complement to calculate the sum of these segments. The calculated sum is then sent along with the data to the receiver. At the receiver's end, the same process is repeated and if all zeroes are obtained in the sum, it means that the data is correct.
    - Checksum - Operation at Sender's Side
      - Firstly, the data is divided into k segments each of m bits.
      - On the sender’s end, the segments are added using 1’s complement arithmetic to get the sum. The sum is complemented to get the checksum.
      - The checksum segment is sent along with the data segments.
    - Checksum - Operation at Receiver's Side
      - At the receiver’s end, all received segments are added using 1’s complement arithmetic to get the sum. The sum is complemented.
      - If the result is zero, the received data is accepted; otherwise discarded.


Original Data
10011001
11100010
00100100
10000100
1
2
3
4
k=4, m=8
Sender
1 10011001
2 11100010
101111011
1
01111100
3 00100100
10100000
4 10000100
100100100
1
Sum: 00100101
CheckSum: 11011010
Reciever
1 10011001
2 11100010
101111011
1
01111100
3 00100100
10100000
4 10000100
100100100
1
00100101
11011010
Sum: 11111111
Complement: 00000000
Conclusion: Accept Data

- 4. Cyclic Redundancy Check (CRC)
  - Unlike the checksum scheme, which is based on addition, CRC is based on binary division.
  - In CRC, a sequence of redundant bits, called cyclic redundancy check bits, are appended to the end of the data unit so that the resulting data unit becomes exactly divisible by a second, predetermined binary number.
  - At the destination, the incoming data unit is divided by the same number. If at this step there is no remainder, the data unit is assumed to be correct and is therefore accepted.
  - A remainder indicates that the data unit has been damaged in transit and therefore must be rejected.

- CRC Working:
  - We have given dataword of length n and divisor of length k.
- Step 1: Append (k-1) zero's to the original message
- Step 2: Perform modulo 2 division
- Step 3: Remainder of division = CRC
- Step 4: Code word = Data with append k-1 zero's + CRC
- Note:
- CRC must be k-1 bits
- Length of Code word = n+k-1 bits

Original message

1010000

Generator polynomial x^3+1

$1x^3+0x^2+0x^1+1x^0$

CRC generator

1001 → 4-bit

If CRC generator is of n bit then append (n-1) zeros in the end of original message

Sender

```
1001 | 1010000000
     @ 1001
       0011000000
     @ 1001
        01019000
     @ 1001
         0011000
     @ 1001
          01010
     @ 1001
           0011
```

Message to be transmitted

```
1010000000
     +011
----------
1010000011
```

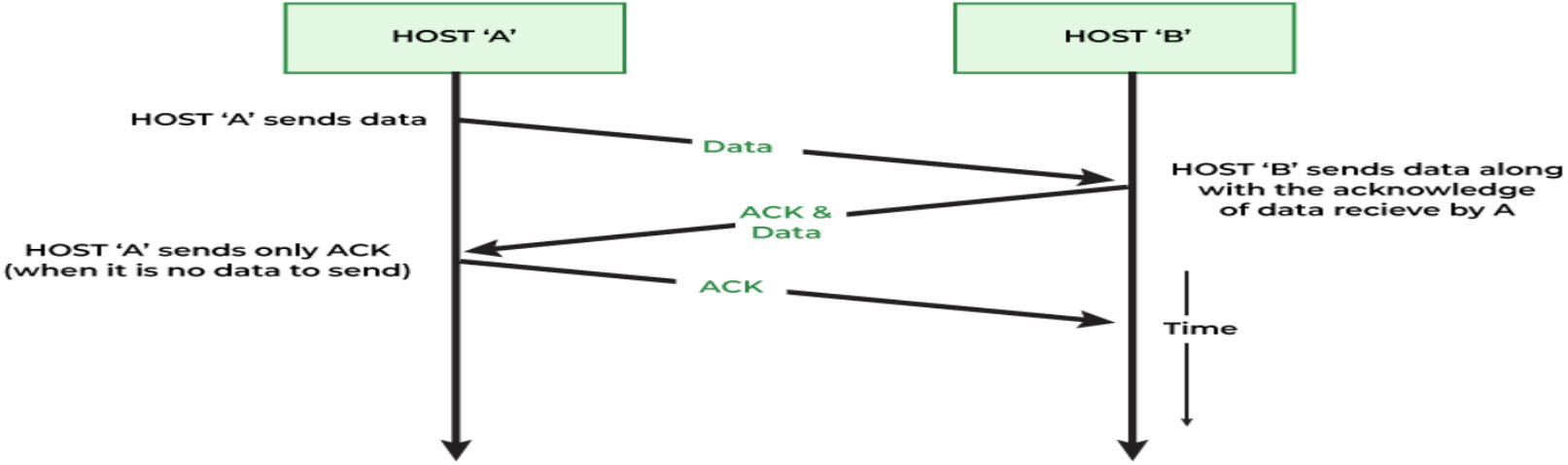
Receiver

```
1001 | 1010000011
     @ 1001
       0011000011
     @ 1001
        01010011
     @ 1001
         0011011
     @ 1001
          01001
     @ 1001
           0000
```

Zero means data is accepted

Piggybacking

- Piggybacking is a technique in computer networks where a device sends an acknowledgment (ACK) for received data within a separate, outgoing data packet instead of sending a dedicated ACK packet. This combines two separate messages into one, reducing network overhead and maximizing bandwidth efficiency, especially in full-duplex communication where data is being sent in both directions.



- How it works
- Data transmission: When Device A sends data to Device B, it waits for a confirmation.
- Delayed acknowledgment: Instead of immediately sending a separate ACK packet back, Device B waits for a short period.
- Packet combination: If Device B has its own data to send to Device A during this waiting period, it includes the ACK for Device A's data within its own outgoing data packet.
- Receipt: Device A receives the single packet and processes both the new data from Device B and the acknowledgment it sent for the original data.

- Key benefits
- Reduces overhead: By combining ACKs with data, piggybacking reduces the total number of packets sent across the network.
- Increases efficiency: It makes better use of network bandwidth that would otherwise be used for separate ACK packets.
- Improves speed: This technique can improve the speed and efficiency of data transmission, particularly in two-way communication.

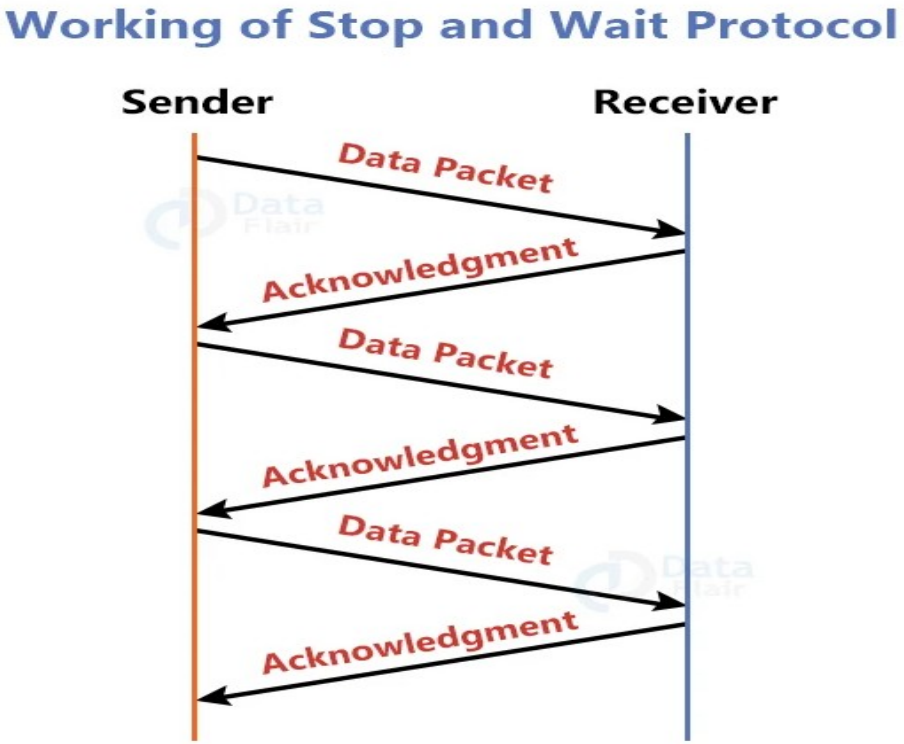
- Example
- Imagine two people, Alex and Ben, are talking on the phone.
- Without piggybacking: Alex finishes a sentence and waits for Ben to say, "Okay, I heard you," before Alex continues talking. If Ben just heard Alex and has nothing to add, he would say, "Okay," and wait for Alex to speak again. This creates a lot of back-and-forth.
- With piggybacking: Alex finishes a sentence. Ben hears it and waits for a moment. If Ben wants to add his own thought, he can say, "Okay, I heard you, and also..." by including the acknowledgment within his own statement. This is more efficient and keeps the conversation flowing.

- What is ARQ (Automatic Repeat Request)?
- ARQ stands for Automatic Repeat Request also known as Automatic Repeat Query. ARQ is an error-control strategy used in a two-way communication system. It is a group of error-control protocols to achieve reliable data transmission over an unreliable source or service. These protocols reside in the Transport Layer and Data Link Layer of the OSI(Open System Interconnection) model. These protocols are responsible for the automatic retransmission of packets that are found to be corrupted or lost during the transmission process.

Simplex Link Layer Protocols

- **Stop and Wait Protocol:** Stop and wait indicates that whatever data the sender wishes to convey is sent to the recipient. After transferring the data, he comes to a halt and waits for the receiver's acknowledgment. The stop and wait protocol is classified as a flow control protocol. Flow control is one of the data connection layer's services.
- Stop and wait protocol is a data-link layer protocol that is used for data transmission across noiseless channels. It supports unidirectional data transmission, which means that it can only send or receive data at the same time. It provides a flow-control mechanism but no error-control mechanism. The reason behind using this frame is that after delivering the frame, the sender waits for the acknowledgment before sending the next frame.

Working of Stop and Wait Protocol:



- The stop and wait technique is depicted in the diagram above. If there is a sender and a receiver, the sender sends the packet, which is referred to as a data packet. The sender will not transmit the second packet until the first packet is acknowledged. The acknowledgment for the data packet received is sent by the receiver.
- The sender transmits the next packet after receiving the acknowledgment. This method is repeated until all of the packets are not transmitted. This protocol’s major virtue is its simplicity, but it also has significant drawbacks.

- **Disadvantages of Stop and Wait Protocol:**

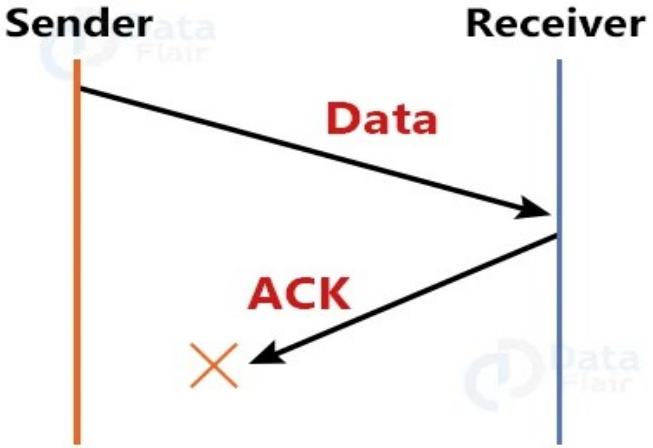
1. Lost Data:

Assume the sender sends the data and it is lost. For a long time, the receiver has been waiting for the data. Because the data is not received by the recipient, no acknowledgment is sent. The sender will not send the following packet since it has not received any acknowledgment. This issue arises as a result of missing data.

This causes 2 problems:

- The sender waits indefinitely for an acknowledgment.
- The receiver waits an indefinite time for data.

2. Lost Acknowledgement:

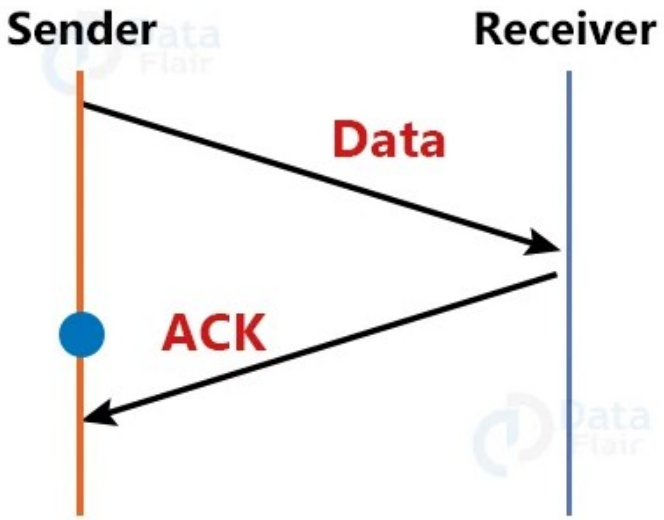


Assume the data is sent by the sender and received by the recipient. When the packet is received, the receiver sends an acknowledgment. In this situation, the acknowledgment is lost in a network, and the sender has no possibility of receiving it. There is also no way for the sender to transmit the following packet because in the stop and wait protocol, the next packet cannot be sent until the preceding packet's acknowledgment is received.

This causes the following problem:

- The sender waits indefinitely for an acknowledgment.

3. Delayed Data or Acknowledgement:

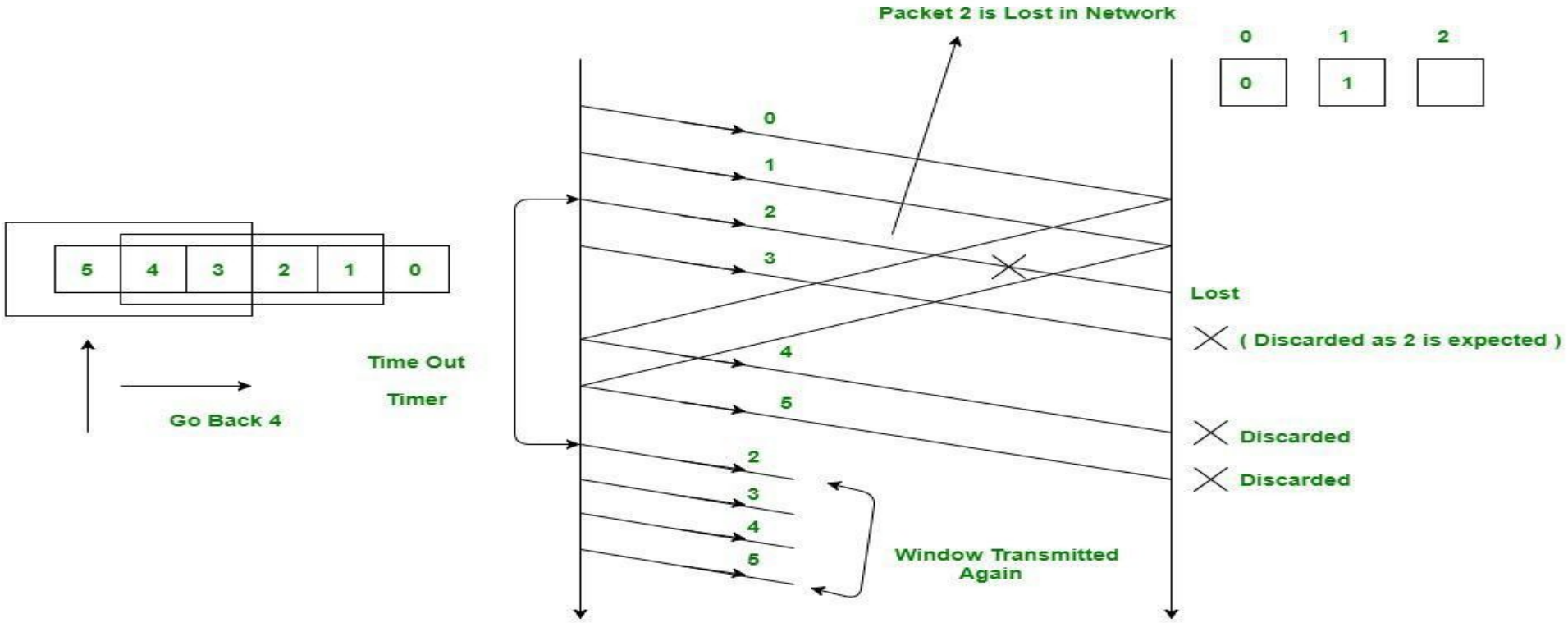


Assume the data is sent by the sender and received by the recipient. The receiver then transmits the acknowledgment, but it is received after the sender's timeout period has expired. Because the acknowledgment is received late, it may be mistaken for the acknowledgment of another data packet.

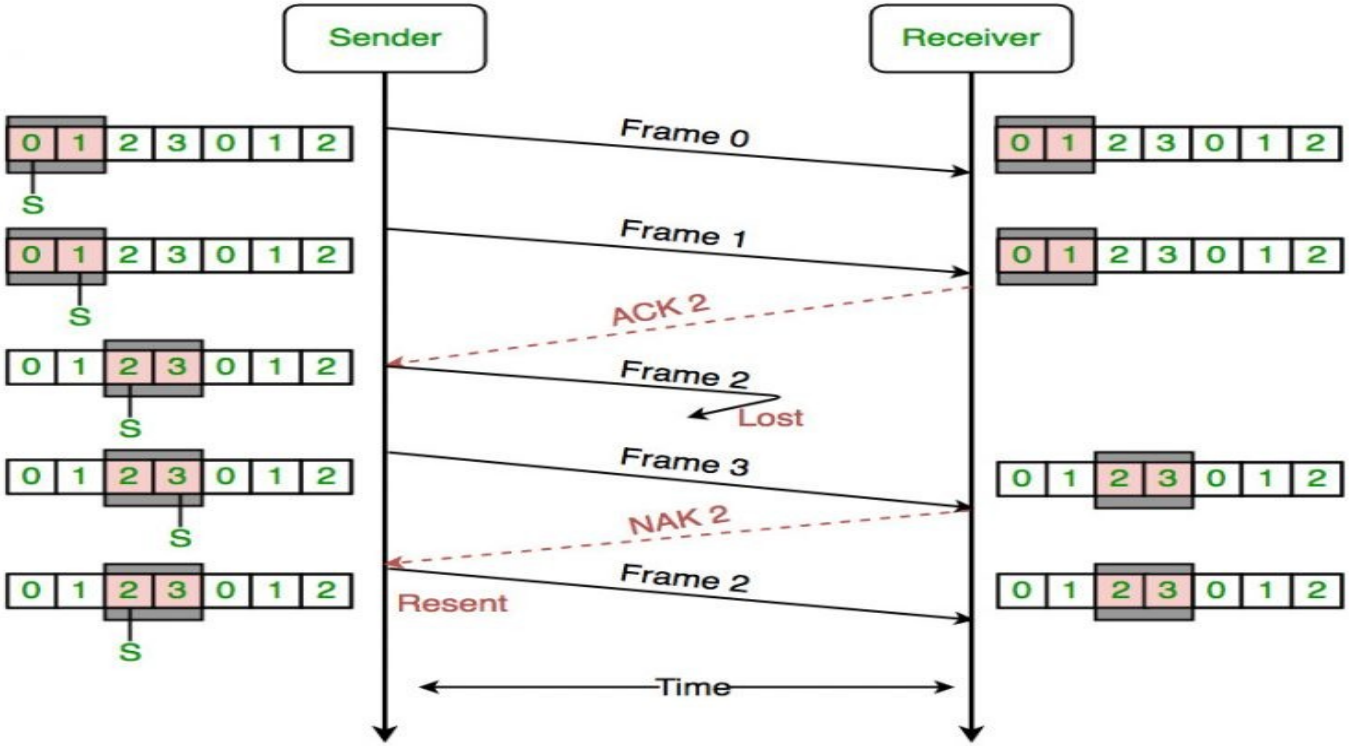
- Sliding Window Protocol
- Sliding Window Protocol allows sending multiple packets before receiving acknowledgments.
- Using a "window" to manage packet flow, it improves network efficiency and throughput, especially over long-distance or high-latency connections.
- Sliding window protocols are used where reliable and ordered delivery of packets is needed, such as in the Data Link Layer and the Transmission Control Protocol in Transport Layer.

- Types of Sliding Window Protocol
- There are two types of Sliding Window Protocol which include Go-Back-N ARQ and Selective Repeat ARQ:
- Go-Back-N ARQ: Multiple frames can be sent before an acknowledgment is received. If a frame is lost or an acknowledgment is missing, all frames from that point onward are retransmitted.
- Selective Repeat ARQ: Multiple frames are sent before acknowledgment, but only the lost or erroneous frames are retransmitted. The receiver can store out-of-order frames until the missing ones arrive.

- **GO-BACK-N ARQ Protocol:**
- The Go-Back-N protocol is a sliding window protocol used for reliable data transfer in computer networks. It is a sender-based protocol that allows the sender to transmit multiple packets without waiting for an acknowledgement for each packet. The receiver sends a cumulative acknowledgement for a sequence of packets, indicating the last correctly received packet.
- If any packet is lost, the receiver sends a negative acknowledgement (NACK) for the lost packet, and the sender retransmits all the packets in the window starting from the lost packet. The sender also maintains a timer for each packet, and if an acknowledgement is not received within the timer's timeout period, the sender retransmits all packets in the window.


Packet 2 is Lost in Network
0
1
2
0
1
5
4
3
2
1
0
Time Out
Timer
Go Back 4
0
1
2
3
4
5
2
3
4
5
Lost
(Discarded as 2 is expected)
Discarded
Discarded
Window Transmitted
Again

- **Selective Repeat Protocol:**
- The Selective Repeat protocol is used for reliable data transfer in computer networks. It is a receiver-based protocol that allows the receiver to acknowledge each packet individually, rather than a cumulative acknowledgement of a sequence of packets. The sender sends packets in a window and waits for acknowledgements for each packet in the window.
- If a packet is lost, the receiver sends a NACK for the lost packet, and the sender retransmits only that packet. The sender also maintains a timer for each packet, and if an acknowledgement is not received within the timer's timeout period, the sender retransmits only that packet.


Sender
Receiver
Frame 0
Frame 1
ACK 2
Frame 2
Lost
Frame 3
NAK 2
Frame 2
Resent
Time
S

Difference Between Go-Back-N and Selective Repeat Protocol

Go-Back-N Protocol	Selective Repeat Protocol
In Go-Back-N Protocol, if the sent frame are find suspected then all the frames are re-transmitted from the lost packet to the last packet transmitted.	In selective Repeat protocol, only those frames are re-transmitted which are found suspected.
Sender window size of Go-Back-N Protocol is N.	Sender window size of selective Repeat protocol is also N.
Receiver window size of Go-Back-N Protocol is 1.	Receiver window size of selective Repeat protocol is N.
Go-Back-N Protocol is less complex.	Selective Repeat protocol is more complex.
In Go-Back-N Protocol, neither sender nor at receiver need sorting.	In selective Repeat protocol, receiver side needs sorting to sort the frames.
In Go-Back-N Protocol, type of Acknowledgement is cumulative.	In Selective Repeat protocol, type of Acknowledgement is individual.
In Go-Back-N Protocol, Out-of-Order packets are NOT Accepted (discarded) and the entire window is re-transmitted.	In selective Repeat protocol, Out-of-Order packets are Accepted.
In Go-Back-N Protocol, if Receives a corrupt packet, then also, the entire window is re-transmitted.	In selective Repeat protocol, if Receives a corrupt packet, it immediately sends a negative acknowledgement and hence only the selective packet is retransmitted.